

Jeffrey Blay

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Professional Summary: Geospatial Data Scientist with 5+ years of experience developing machine learning and deep learning models for large-scale geospatial datasets. Passionate about applying data science and geospatial intelligence to support evidence-based decisions, address complex societal challenges, and improve quality of life through data-informed solutions. Skilled in building end-to-end data pipelines and spatial AI systems using various programming languages and geospatial tools.

Certifications

- IBM Professional Data Engineering Certificate — Coursera (In Progress, Expected May 2026)

Education

North Carolina A&T State University, Ph.D. Applied Science and Technology (Expected December 2026)

- Specialization: Geospatial Data Science
- Relevant Courses: Machine Learning and Data Mining, Big Data Analytics, Advanced Geospatial Analysis, Neural networks, Statistical Methods, Multivariate Statistics for Engineers

Yale School of the Environment, Master of Environmental Science (August 2021 – May 2023)

- Specialization: Environmental Data science for Urban sustainability and resilience
- Relevant Courses: Advanced Urban Remote Sensing, SDGs & Implementation, Statistics for Environmental Sciences, Data Exploration & Analysis, Environmental Data Science, Introductory Machine Learning

Professional Experience

Remote Sensing & GIS Lab, North Carolina A&T State University - Greensboro, North Carolina-USA.

Graduate Research Assistant, NASA Flood Project | Part-Time

September 2025 - Present

- Implementing CNN (UNet, Attention-UNet) and transformer (Swin-UNet, SegFormer) models for urban flood-depth prediction in PyTorch.
- Designing a physics-informed deep learning framework to integrate hydrostatic constraints; running controlled experiments benchmarking physics-informed vs. standard deep learning models.
- Building end-to-end full-stack interactive dashboard deployment framework with FastAPI and interactive dashboards for stakeholder flood risk and vulnerability assessment.

NASA DEAP Institute, North Carolina A&T State University - Greensboro, North Carolina-USA.

Data Science Research Assistant, NASA Data Project | Full-Time

June - August 2025

- Trained and evaluated deep learning models (U-Net, Attention U-Net, U-Net++) using TensorFlow for urban flood-depth prediction, producing visualizations (ArcGIS Online) to inform flood-risk assessment.
- Developed and published ezprocess, an open-source Python library for geospatial data preprocessing (PyPI/GitHub), reduced data preparation time by 70% across ML/DL workflows.
- Managed codes, experiments, and library version control using Git and GitHub, enabling collaborative development across research teams.

Remote Sensing & GIS Lab, North Carolina A&T State University- Greensboro, North Carolina-USA.

Graduate Research Assistant, NASA Data Project | Part-Time

September 2024 – May 2025

- Engineered geospatial ETL pipelines in ArcGIS and Python (rasterio, ArcPy) to ingest, preprocess, and integrate multi-source geospatial datasets for ML training.
- Led 3-person team to design, build, and publish a public benchmark dataset, managing data schema, QA, and documentation from end to end.
- Implemented U-Net and pix2pix cGAN workflows to infer flood depth from UAV imagery and LiDAR point clouds.

GEMS Institute, North Carolina A&T State University - Greensboro, North Carolina-USA.

Data Science Research Assistant, NSF Multi Modal Data Fusion Project | Full-Time

June - August 2024

- Processed 700k+ 1-m resolution geospatial pixels via NumPy and scikit-learn pipelines for large-scale flood prediction with machine learning models (Random Forest & XGboost).
- Collaborated with a 5-member team to develop and publish a multi-source geospatial dataset for flood inundation modeling; designed automated pipelines preprocess and integrate multi-sensor datasets using ArcPy in ArcGIS Pro.

- Completed training workshop in cloud-based data storage management and retrieval using Amazon S3.

Remote Sensing & GIS Lab, North Carolina A&T State University- Greensboro, North Carolina-USA.

Graduate Research Assistant, NOAA Flood Project | Part-Time

September 2023 – May 2024

- Engineered web-scraping workflows with python (Beautiful Soup) to curate a custom image classification training datasets.
- Built a framework using PyTorch to train and evaluate image classification models (GoogleNet, ResNet18) for automated flood classification of crowdsourced images.
- Led a technical assessment of GAN-based approaches for flood mapping using remote sensing.

Urban Resources Initiative, Non-Governmental Organization – New Haven, Connecticut-USA.

Community Group Manager, Summer Greenspace Community Initiative | Full-Time

May - August 2023

- Supervised and 7 community greenspace groups, supporting planning and implementation of local green initiatives.
- Organized weekly project planning sessions and coordinated strategies to advance community sustainability and greenspace development goals.
- Collaborated with community stakeholders to design and visualize project concepts using ArcGIS Pro and QGIS.

Hixon Center For Urban Ecology, Yale University - New Haven, Connecticut-USA.

Research Assistant, NASA Environmental Justice Project | Part-Time

September 2022 – May 2023

- Utilized multi-spectral satellite imagery and geostatistical techniques to conduct census block-level analysis to identify UHI-vulnerable urban communities.
- Conducted a block level analysis of Surface temperature using ArcPy and ArcGIS Pro SQL query builder.
- Used R (sp, dplyr, ggplot2) for time-series statistical analysis and mapping of urban heat and tree cover gaps.

Seto Lab, Yale University - New Haven, Connecticut-USA.

Geospatial Research Assistant, Urban Africa Project | Part-Time

September 2021 - May 2022

- Developed and optimized Google Earth Engine workflows for large-scale satellite data acquisition and preprocessing.
- Trained satellite image segmentation models in Python on a linux-HPC, implementing scalable batch processing pipelines.
- Built custom R and Python scripts for geospatial data processing, analysis, and visualizations.

Ghana Statistical Service, National Government Agency – Accra, Ghana

GIS Data Officer, National Census Secretariat | Full-Time

September 2020 – June 2021

- Validated and corrected large-scale census geospatial datasets as part of a 15-member nationwide data operations team.
- Produced daily near-real-time spatial analytics dashboards, monitoring field census operations across the country.

Projects

Weather Alert Dashboard [Live](#) Python · FastAPI · JavaScript · Docker · CI/CD · REST API · GitHub Pages

2026

- Built real-time weather monitoring platform integrating National Weather Service API; REST API with filtering, analytics, and risk-scoring across all 50 U.S. states.

Ezprocess - Open-Source Python Library Python · PyPI · GitHub · Geospatial ETL

2025

- Designed, packaged, and published geospatial data preprocessing library to PyPI/GitHub; reduces preparation time by 70% .

Technical Skills

Programming Languages: Python, R, SQL, JavaScript

Geospatial & Remote Sensing: ArcGIS Suite, QGIS, Google Earth Engine, ArcPy, GDAL, ENVI, SNAP

ML/ DL Frameworks: PyTorch, TensorFlow, Scikit-Learn, Computer vision, Physics-Informed Neural Networks (PINNs)

Data Engineering & Processing: ETL Pipelines, Pandas, RDBMS, Web Scraping, Geospatial Data Processing

Visualization & Decision Support: Matplotlib, Seaborn, Dashboards, ArcGIS StoryMaps, Power BI, R Shiny (R), Tableau

Web, APIs & Applications: FastAPI, RESTful APIs, HTML/CSS, Streamlit

Cloud, DevOps & Infrastructure: AWS S3, Docker, CI/CD, Git, GitHub, HPC, GitHub Pages, Render

Databases: PostgreSQL, MySQL, IBM DB2